Result

1 Project information

The basic information of the project is shown below:

- Project ID: F23A480000634-01_ORYaozuO
- Product name: filter
- Sample size: 20
- Library type: rRNA Depletion stranded Library
- Sequencing Platform: DNBseq
- Sequencing read Length: PE150
- Clean fastq phred quality score encoding:Phred+33

2 Data production

After sequencing, the raw reads were filtered. Data filtering includes removing adaptor sequences, contamination and low-quality reads from raw reads.

Table 1 Statistics of clean data

Sample Name	Clean Reads	Clean Base	Read Length	Q20(%)	Q30(%)	GC(%)
C4	24,078,883	7,223,664,900	PE150	97.66	91.41	48.94
C5	24,139,798	7,241,939,400	PE150	97.93	92.48	50.28
C6	24,098,436	7,229,530,800	PE150	97.75	91.63	48.79
EE24	24,088,519	7,226,555,700	PE150	97.90	92.52	50.50
EE26	24,104,030	7,231,209,000	PE150	97.88	92.20	49.28
GenX4	24,111,741	7,233,522,300	PE150	97.86	92.60	54.13
GenX5	24,119,017	7,235,705,100	PE150	97.91	92.33	49.14
GenX6	24,083,914	7,225,174,200	PE150	97.84	92.14	48.98
PFBS4	24,074,785	7,222,435,500	PE150	97.77	91.75	49.80
PFBS5	24,125,559	7,237,667,700	PE150	97.18	91.29	64.34
PFBS6	24,077,332	7,223,199,600	PE150	97.94	92.47	50.33
PFHxS4	24,038,429	7,211,528,700	PE150	97.74	91.68	49.35
PFHxS5	24,058,666	7,217,599,800	PE150	97.93	92.55	50.58
PFHxS6	24,084,732	7,225,419,600	PE150	97.78	92.13	52.00
PFOA4	24,070,008	7,221,002,400	PE150	98.01	92.95	51.94
PFOA5	24,017,741	7,205,322,300	PE150	97.83	91.98	49.29
PFOA6	24,043,777	7,213,133,100	PE150	97.84	92.07	49.24
PFOS4	24,067,080	7,220,124,000	PE150	97.89	92.22	49.61
PFOS5	24,086,298	7,225,889,400	PE150	97.95	92.37	49.23
PFOS6	24,062,853	7,218,855,900	PE150	97.87	92.16	50.18

- Sample Name：Sample Name
- Clean Reads：Clean reads
- Clean Base：Clean bases
- Read Length：Read length
- Q20(%)：Proportion of Q20
- Q30(%)：Proportion of Q30
- GC(%)：Proportion of GC

3 Quality control

The quality of data was examined after filtering.

3.1 The distribution of base percentage and qualities along reads after data filtering

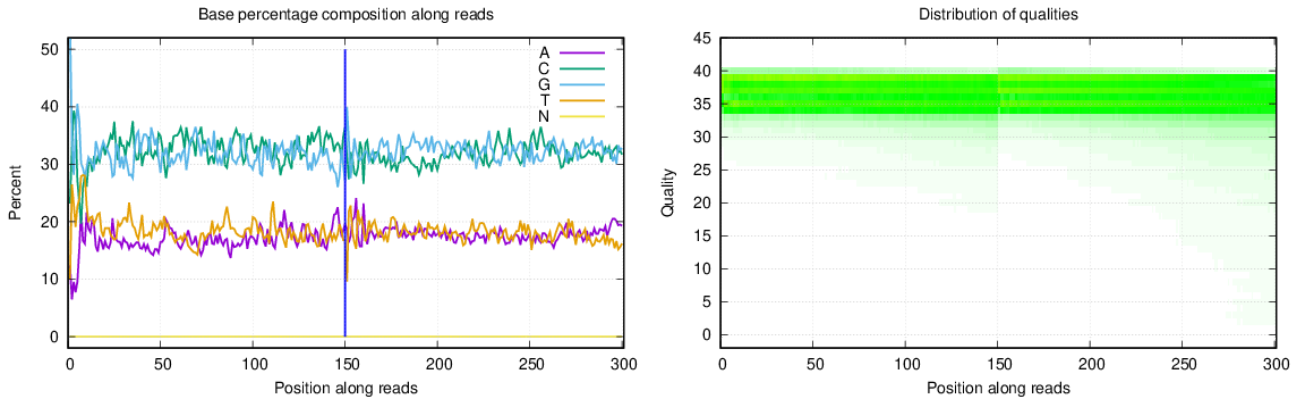


Figure 1 PFBS5

In the left figure, x-axis represents base position along reads, y-axis represents base percentage at the position; each color represents a type of nucleotide. Under normal conditions, the sample does not have AT/GC separation. It is normal to see fluctuations in the first several bp positions, which is caused by random primer and the instability of enzyme-substrate binding at the beginning of the sequencing reaction. In the right figure, x-axis represents base position along reads, y-axis represents base quality; each dot represents the base quality of the corresponding position along reads, color intensity reflects the number of nucleotides, a more intense color along a quality value indicates a higher proportion of this quality in the sequencing data.

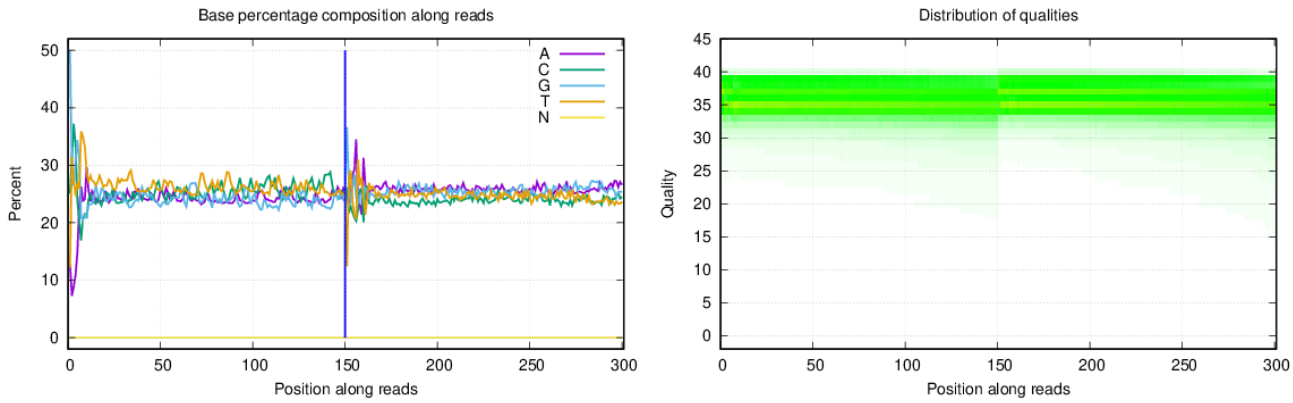


Figure 2 PFOS4

In the left figure, x-axis represents base position along reads, y-axis represents base percentage at the position; each color represents a type of nucleotide. Under normal conditions, the sample does not have AT/GC separation. It is normal to see fluctuations in the first several bp positions, which is caused by random primer and the instability of enzyme-substrate binding at the beginning of the sequencing reaction. In the right figure, x-axis represents base position along reads, y-axis represents base quality; each dot represents the base quality of the corresponding position along reads, color intensity reflects the number of nucleotides, a more intense color along a quality value indicates a higher proportion of this quality in the sequencing data.

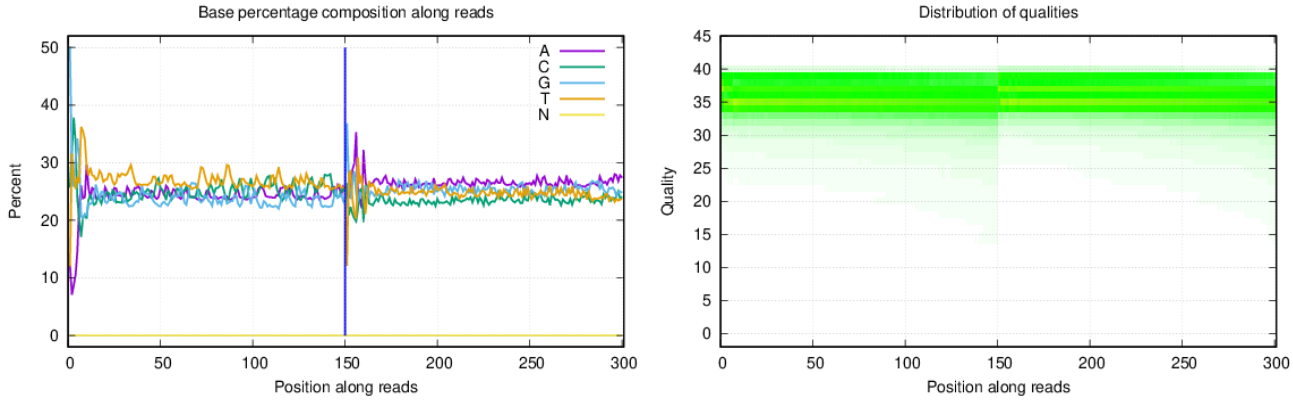


Figure 3 C4

In the left figure, x-axis represents base position along reads, y-axis represents base percentage at the position; each color represents a type of nucleotide. Under normal conditions, the sample does not have AT/GC separation. It is normal to see fluctuations in the first several bp positions, which is caused by random primer and the instability of enzyme-substrate binding at the beginning of the sequencing reaction. In the right figure, x-axis represents base position along reads, y-axis represents base quality; each dot represents the base quality of the corresponding position along reads, color intensity reflects the number of nucleotides, a more intense color along a quality value indicates a higher proportion of this quality in the sequencing data.

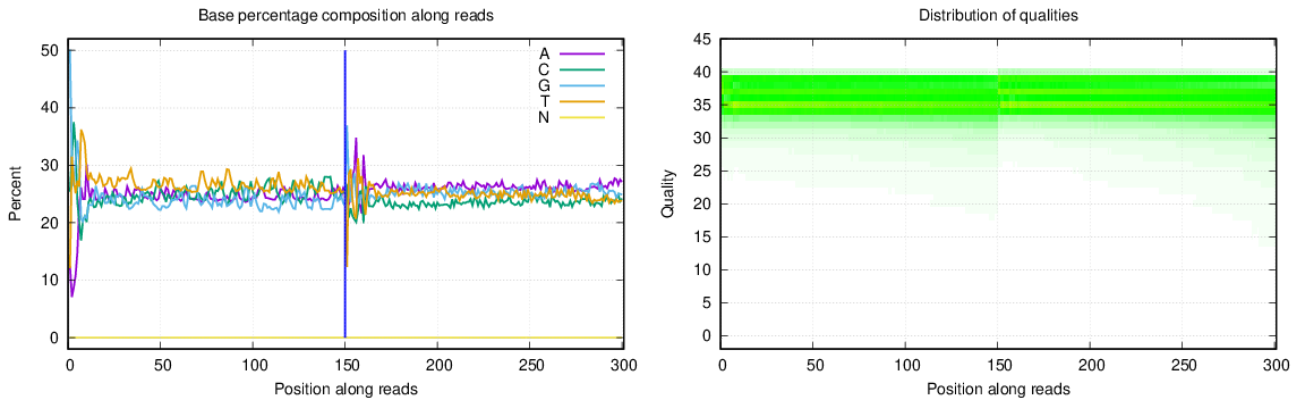


Figure 4 GenX6

In the left figure, x-axis represents base position along reads, y-axis represents base percentage at the position; each color represents a type of nucleotide. Under normal conditions, the sample does not have AT/GC separation. It is normal to see fluctuations in the first several bp positions, which is caused by random primer and the instability of enzyme-substrate binding at the beginning of the sequencing reaction. In the right figure, x-axis represents base position along reads, y-axis represents base quality; each dot represents the base quality of the corresponding position along reads, color intensity reflects the number of nucleotides, a more intense color along a quality value indicates a higher proportion of this quality in the sequencing data.

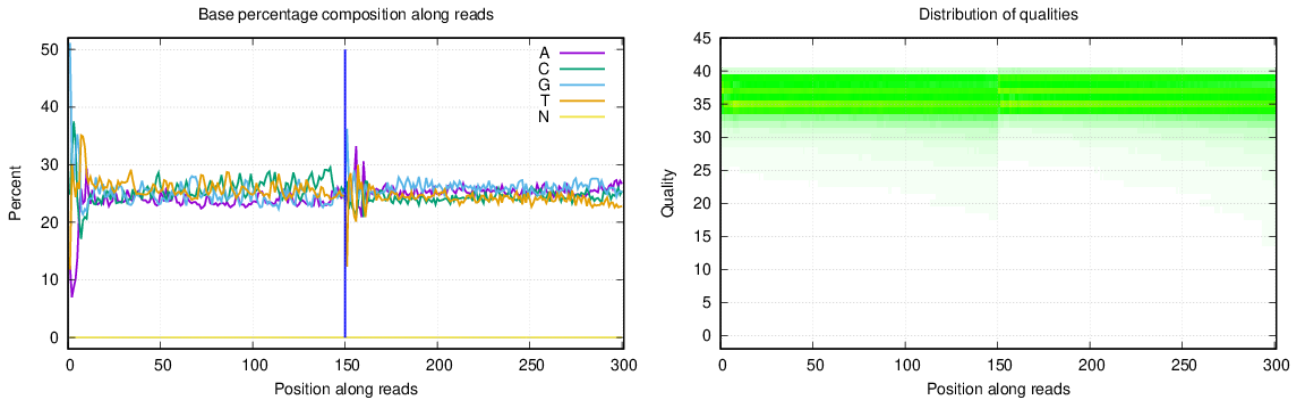


Figure 5 PFHxS5

In the left figure, x-axis represents base position along reads, y-axis represents base percentage at the position; each color represents a type of nucleotide. Under normal conditions, the sample does not have AT/GC separation. It is normal to see fluctuations in the first several bp positions, which is caused by random primer and the instability of enzyme-substrate binding at the beginning of the sequencing reaction. In the right figure, x-axis represents base position along reads, y-axis represents base quality; each dot represents the base quality of the corresponding position along reads, color intensity reflects the number of nucleotides, a more intense color along a quality value indicates a higher proportion of this quality in the sequencing data.

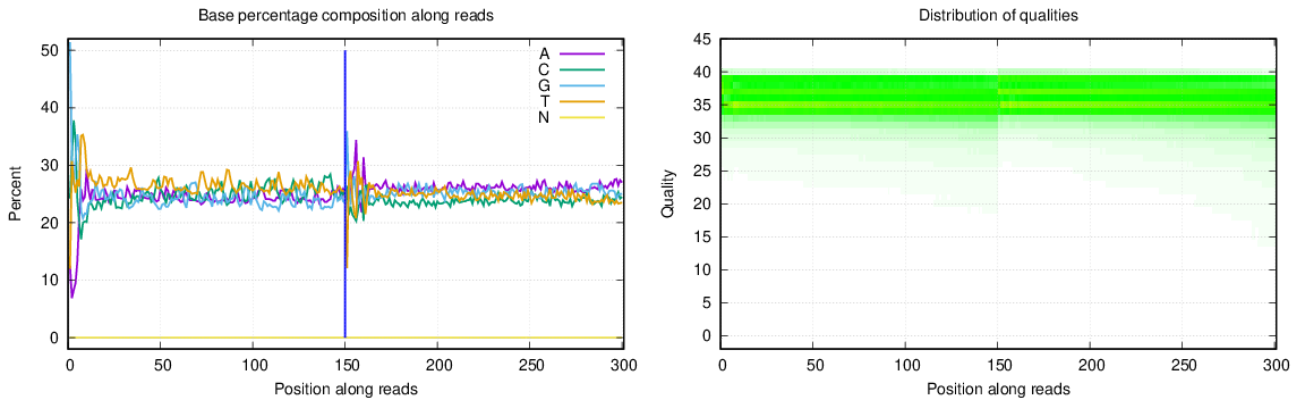


Figure 6 EE26

In the left figure, x-axis represents base position along reads, y-axis represents base percentage at the position; each color represents a type of nucleotide. Under normal conditions, the sample does not have AT/GC separation. It is normal to see fluctuations in the first several bp positions, which is caused by random primer and the instability of enzyme-substrate binding at the beginning of the sequencing reaction. In the right figure, x-axis represents base position along reads, y-axis represents base quality; each dot represents the base quality of the corresponding position along reads, color intensity reflects the number of nucleotides, a more intense color along a quality value indicates a higher proportion of this quality in the sequencing data.

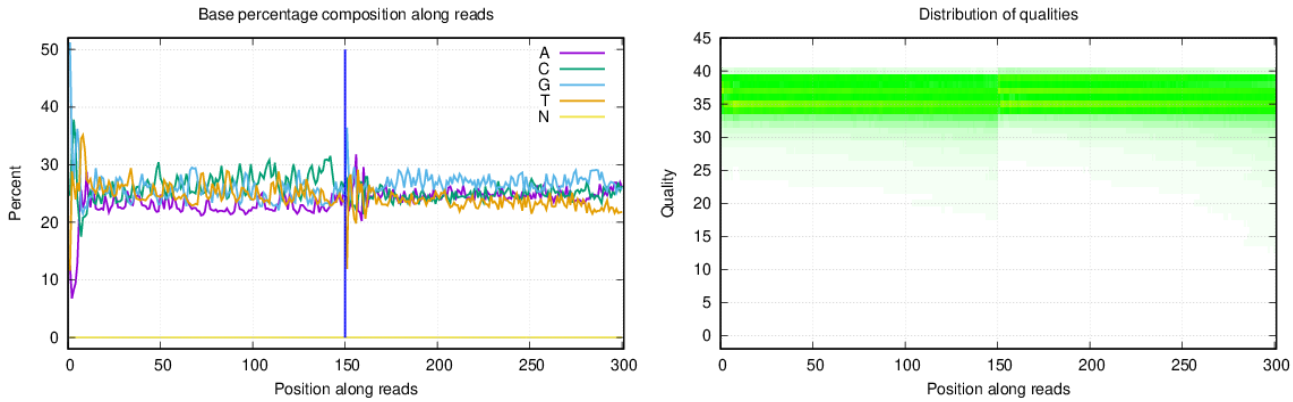


Figure 7 PFHxS6

In the left figure, x-axis represents base position along reads, y-axis represents base percentage at the position; each color represents a type of nucleotide. Under normal conditions, the sample does not have AT/GC separation. It is normal to see fluctuations in the first several bp positions, which is caused by random primer and the instability of enzyme-substrate binding at the beginning of the sequencing reaction. In the right figure, x-axis represents base position along reads, y-axis represents base quality; each dot represents the base quality of the corresponding position along reads, color intensity reflects the number of nucleotides, a more intense color along a quality value indicates a higher proportion of this quality in the sequencing data.

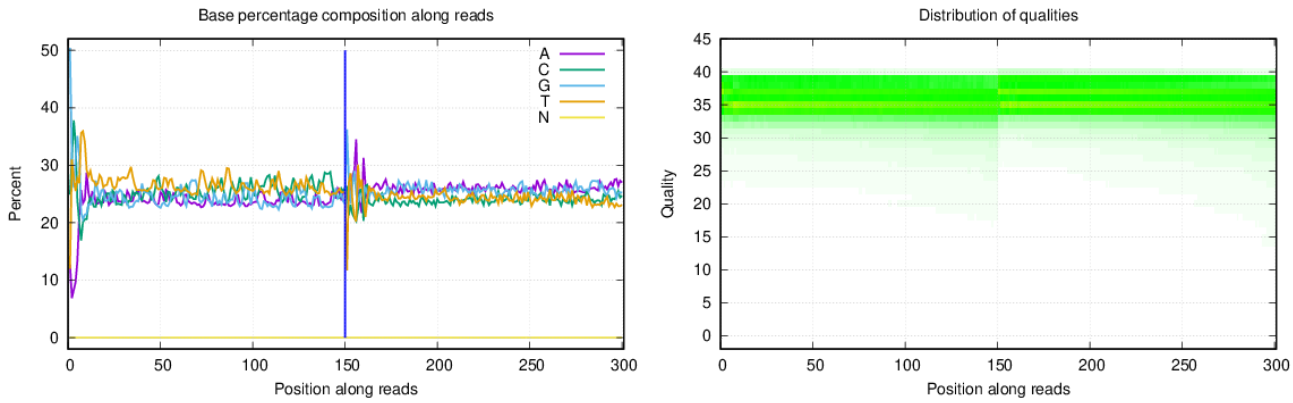


Figure 8 PFBS4

In the left figure, x-axis represents base position along reads, y-axis represents base percentage at the position; each color represents a type of nucleotide. Under normal conditions, the sample does not have AT/GC separation. It is normal to see fluctuations in the first several bp positions, which is caused by random primer and the instability of enzyme-substrate binding at the beginning of the sequencing reaction. In the right figure, x-axis represents base position along reads, y-axis represents base quality; each dot represents the base quality of the corresponding position along reads, color intensity reflects the number of nucleotides, a more intense color along a quality value indicates a higher proportion of this quality in the sequencing data.

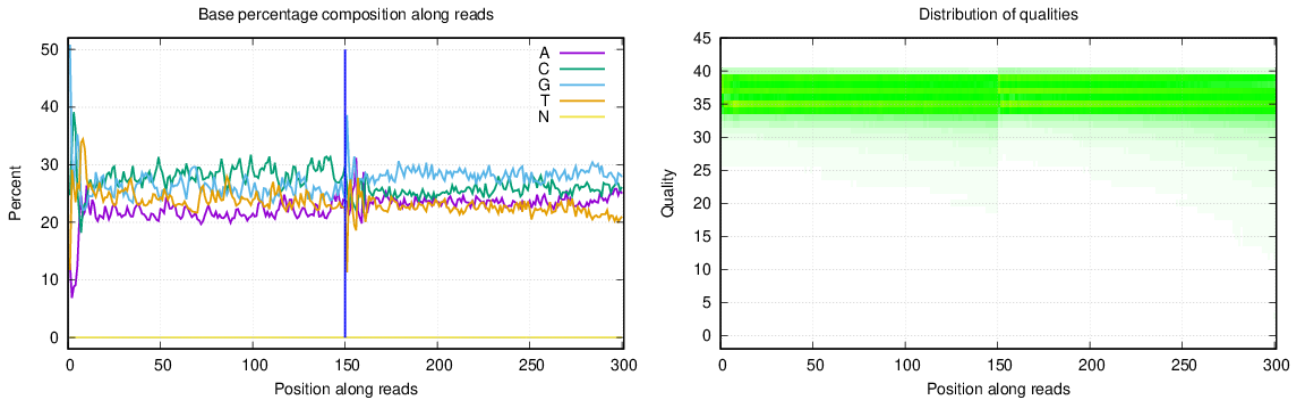


Figure 9 GenX4

In the left figure, x-axis represents base position along reads, y-axis represents base percentage at the position; each color represents a type of nucleotide. Under normal conditions, the sample does not have AT/GC separation. It is normal to see fluctuations in the first several bp positions, which is caused by random primer and the instability of enzyme-substrate binding at the beginning of the sequencing reaction. In the right figure, x-axis represents base position along reads, y-axis represents base quality; each dot represents the base quality of the corresponding position along reads, color intensity reflects the number of nucleotides, a more intense color along a quality value indicates a higher proportion of this quality in the sequencing data.

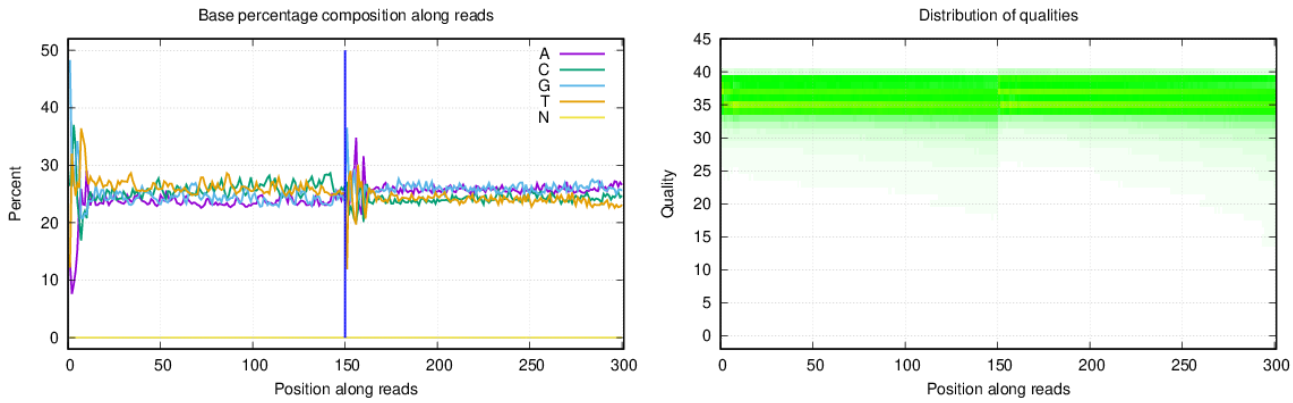


Figure 10 C5

In the left figure, x-axis represents base position along reads, y-axis represents base percentage at the position; each color represents a type of nucleotide. Under normal conditions, the sample does not have AT/GC separation. It is normal to see fluctuations in the first several bp positions, which is caused by random primer and the instability of enzyme-substrate binding at the beginning of the sequencing reaction. In the right figure, x-axis represents base position along reads, y-axis represents base quality; each dot represents the base quality of the corresponding position along reads, color intensity reflects the number of nucleotides, a more intense color along a quality value indicates a higher proportion of this quality in the sequencing data.

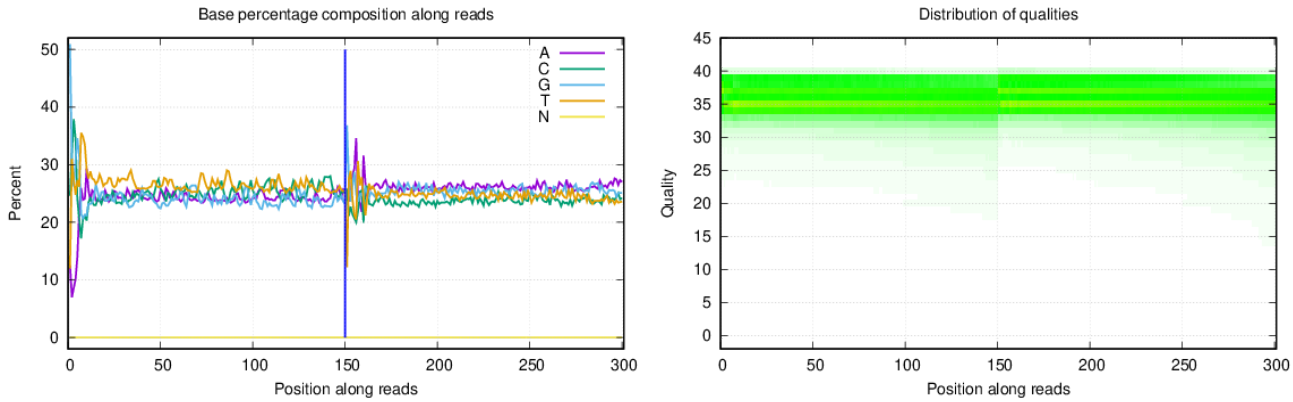


Figure 11 PFHxS4

In the left figure, x-axis represents base position along reads, y-axis represents base percentage at the position; each color represents a type of nucleotide. Under normal conditions, the sample does not have AT/GC separation. It is normal to see fluctuations in the first several bp positions, which is caused by random primer and the instability of enzyme-substrate binding at the beginning of the sequencing reaction. In the right figure, x-axis represents base position along reads, y-axis represents base quality; each dot represents the base quality of the corresponding position along reads, color intensity reflects the number of nucleotides, a more intense color along a quality value indicates a higher proportion of this quality in the sequencing data.

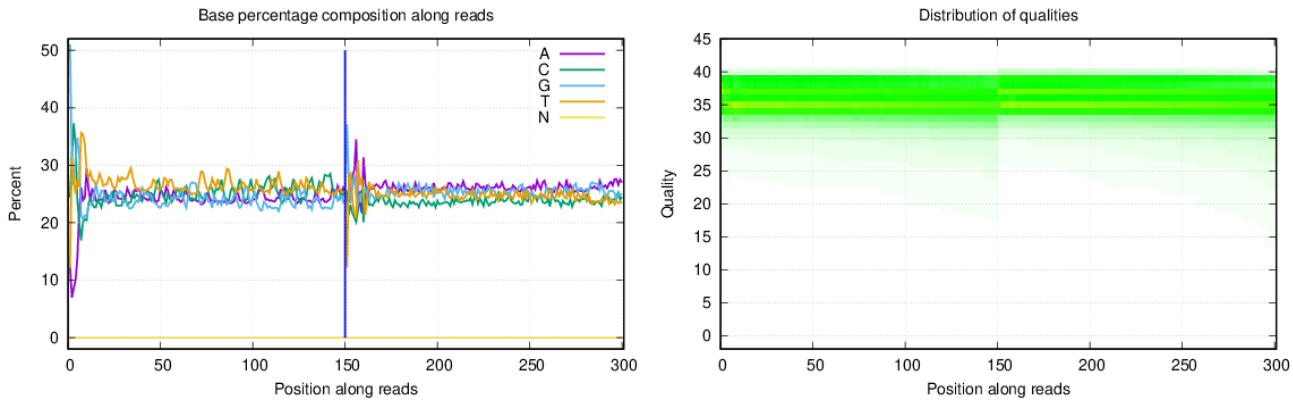


Figure 12 PFOA5

In the left figure, x-axis represents base position along reads, y-axis represents base percentage at the position; each color represents a type of nucleotide. Under normal conditions, the sample does not have AT/GC separation. It is normal to see fluctuations in the first several bp positions, which is caused by random primer and the instability of enzyme-substrate binding at the beginning of the sequencing reaction. In the right figure, x-axis represents base position along reads, y-axis represents base quality; each dot represents the base quality of the corresponding position along reads, color intensity reflects the number of nucleotides, a more intense color along a quality value indicates a higher proportion of this quality in the sequencing data.

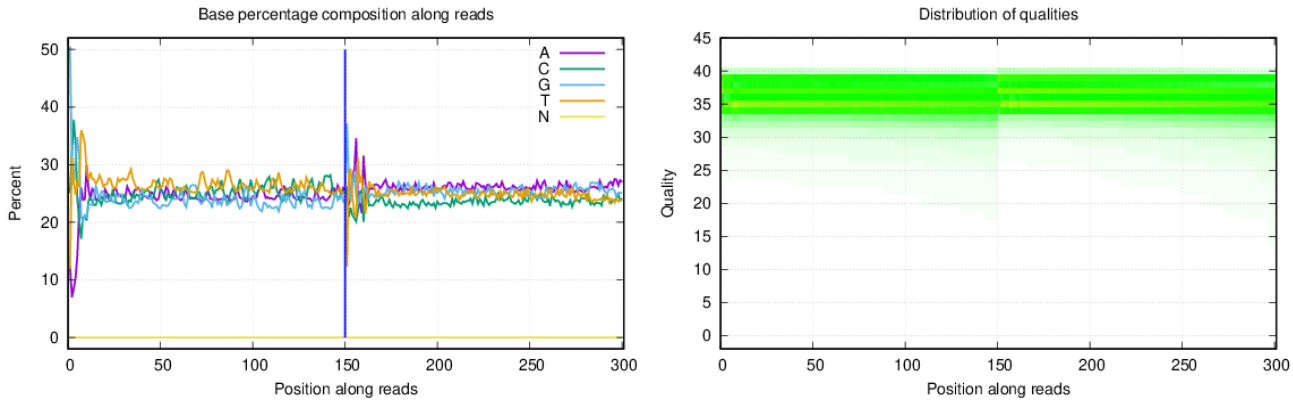


Figure 13 GenX5

In the left figure, x-axis represents base position along reads, y-axis represents base percentage at the position; each color represents a type of nucleotide. Under normal conditions, the sample does not have AT/GC separation. It is normal to see fluctuations in the first several bp positions, which is caused by random primer and the instability of enzyme-substrate binding at the beginning of the sequencing reaction. In the right figure, x-axis represents base position along reads, y-axis represents base quality; each dot represents the base quality of the corresponding position along reads, color intensity reflects the number of nucleotides, a more intense color along a quality value indicates a higher proportion of this quality in the sequencing data.

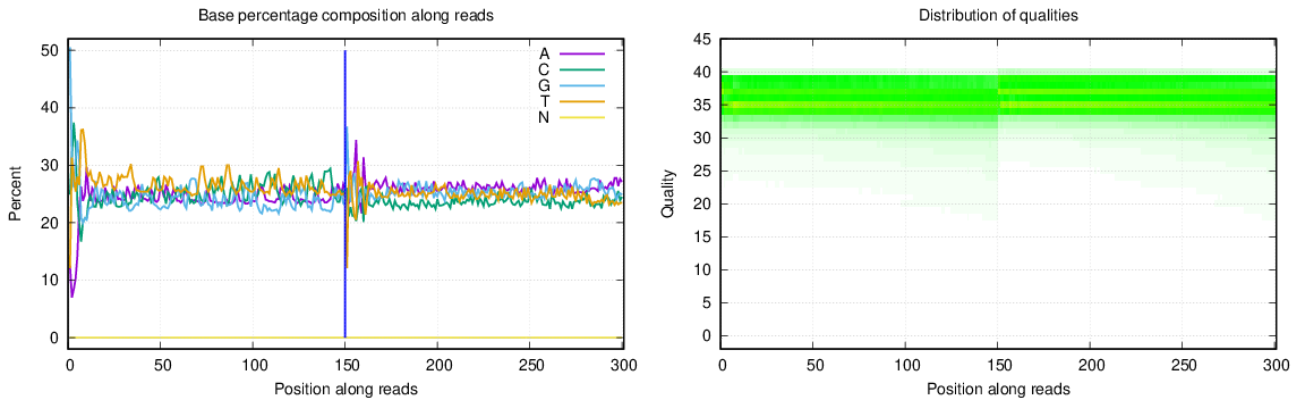


Figure 14 PFOS5

In the left figure, x-axis represents base position along reads, y-axis represents base percentage at the position; each color represents a type of nucleotide. Under normal conditions, the sample does not have AT/GC separation. It is normal to see fluctuations in the first several bp positions, which is caused by random primer and the instability of enzyme-substrate binding at the beginning of the sequencing reaction. In the right figure, x-axis represents base position along reads, y-axis represents base quality; each dot represents the base quality of the corresponding position along reads, color intensity reflects the number of nucleotides, a more intense color along a quality value indicates a higher proportion of this quality in the sequencing data.

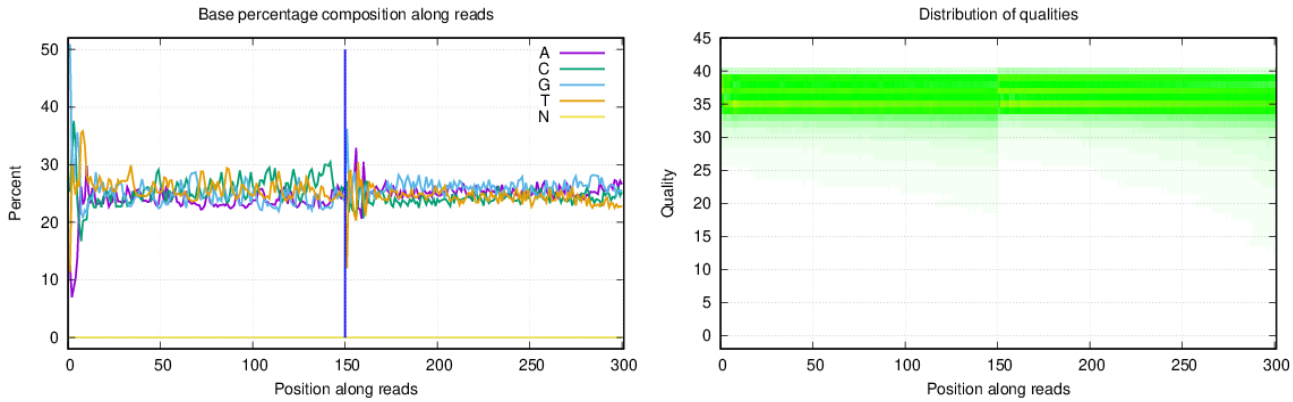


Figure 15 EE24

In the left figure, x-axis represents base position along reads, y-axis represents base percentage at the position; each color represents a type of nucleotide. Under normal conditions, the sample does not have AT/GC separation. It is normal to see fluctuations in the first several bp positions, which is caused by random primer and the instability of enzyme-substrate binding at the beginning of the sequencing reaction. In the right figure, x-axis represents base position along reads, y-axis represents base quality; each dot represents the base quality of the corresponding position along reads, color intensity reflects the number of nucleotides, a more intense color along a quality value indicates a higher proportion of this quality in the sequencing data.

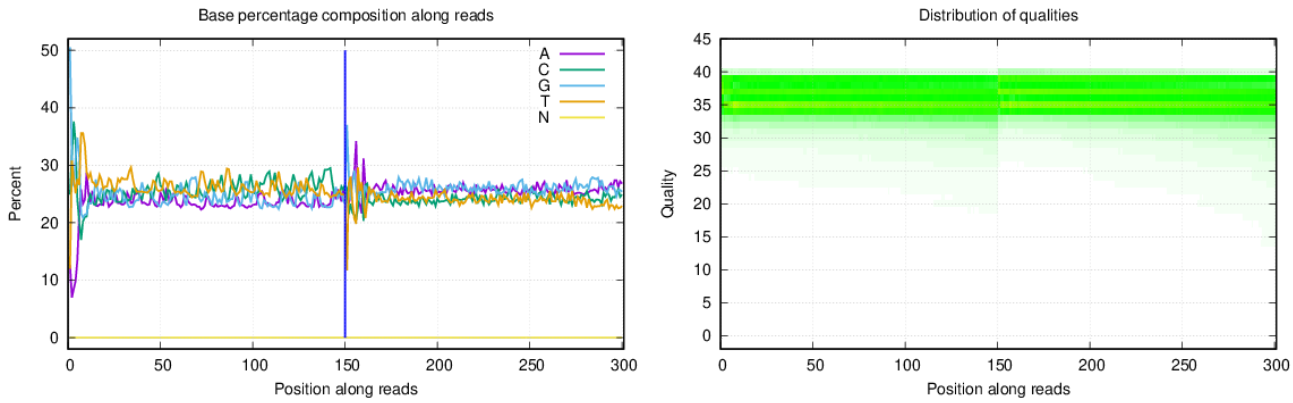


Figure 16 PFBS6

In the left figure, x-axis represents base position along reads, y-axis represents base percentage at the position; each color represents a type of nucleotide. Under normal conditions, the sample does not have AT/GC separation. It is normal to see fluctuations in the first several bp positions, which is caused by random primer and the instability of enzyme-substrate binding at the beginning of the sequencing reaction. In the right figure, x-axis represents base position along reads, y-axis represents base quality; each dot represents the base quality of the corresponding position along reads, color intensity reflects the number of nucleotides, a more intense color along a quality value indicates a higher proportion of this quality in the sequencing data.

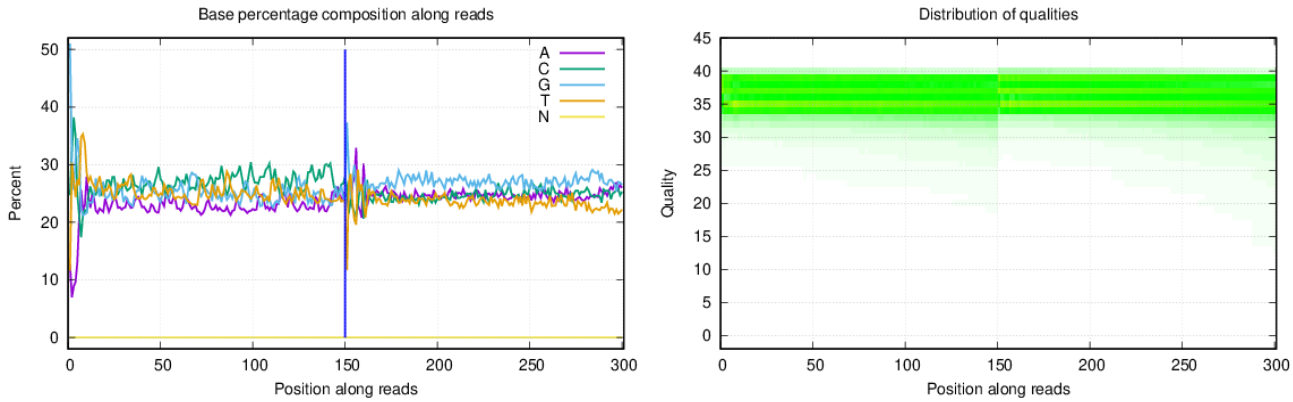


Figure 17 PFOA4

In the left figure, x-axis represents base position along reads, y-axis represents base percentage at the position; each color represents a type of nucleotide. Under normal conditions, the sample does not have AT/GC separation. It is normal to see fluctuations in the first several bp positions, which is caused by random primer and the instability of enzyme-substrate binding at the beginning of the sequencing reaction. In the right figure, x-axis represents base position along reads, y-axis represents base quality; each dot represents the base quality of the corresponding position along reads, color intensity reflects the number of nucleotides, a more intense color along a quality value indicates a higher proportion of this quality in the sequencing data.

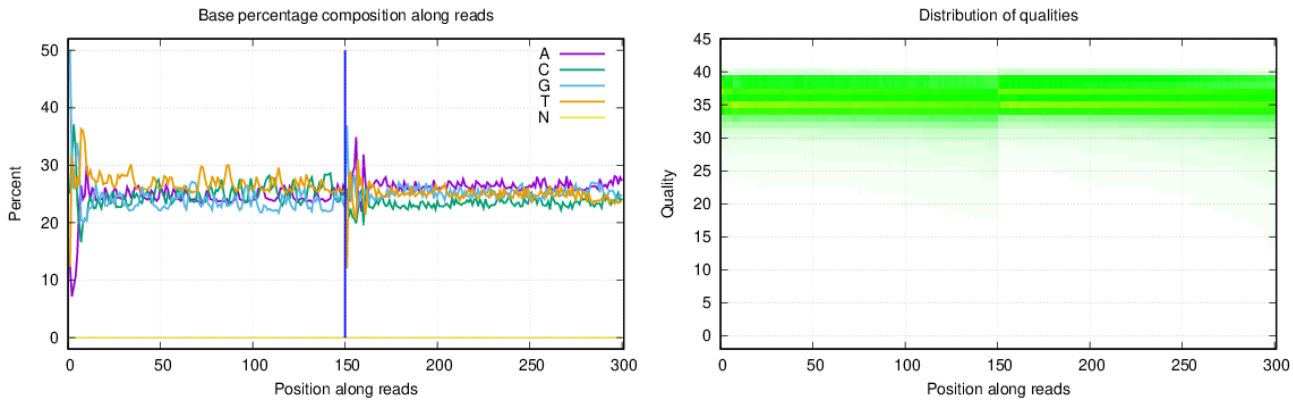


Figure 18 C6

In the left figure, x-axis represents base position along reads, y-axis represents base percentage at the position; each color represents a type of nucleotide. Under normal conditions, the sample does not have AT/GC separation. It is normal to see fluctuations in the first several bp positions, which is caused by random primer and the instability of enzyme-substrate binding at the beginning of the sequencing reaction. In the right figure, x-axis represents base position along reads, y-axis represents base quality; each dot represents the base quality of the corresponding position along reads, color intensity reflects the number of nucleotides, a more intense color along a quality value indicates a higher proportion of this quality in the sequencing data.

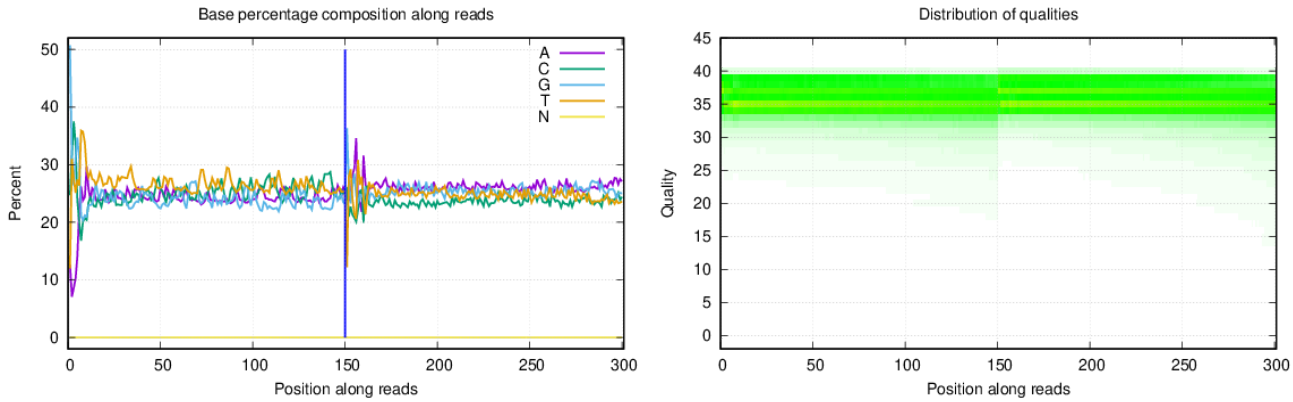


Figure 19 PFOA6

In the left figure, x-axis represents base position along reads, y-axis represents base percentage at the position; each color represents a type of nucleotide. Under normal conditions, the sample does not have AT/GC separation. It is normal to see fluctuations in the first several bp positions, which is caused by random primer and the instability of enzyme-substrate binding at the beginning of the sequencing reaction. In the right figure, x-axis represents base position along reads, y-axis represents base quality; each dot represents the base quality of the corresponding position along reads, color intensity reflects the number of nucleotides, a more intense color along a quality value indicates a higher proportion of this quality in the sequencing data.

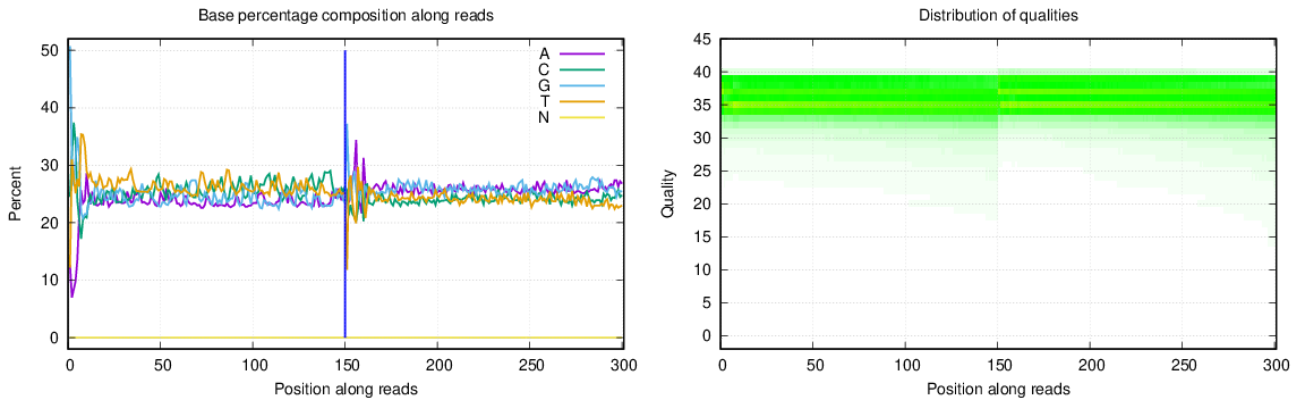


Figure 20 PFOS6

In the left figure, x-axis represents base position along reads, y-axis represents base percentage at the position; each color represents a type of nucleotide. Under normal conditions, the sample does not have AT/GC separation. It is normal to see fluctuations in the first several bp positions, which is caused by random primer and the instability of enzyme-substrate binding at the beginning of the sequencing reaction. In the right figure, x-axis represents base position along reads, y-axis represents base quality; each dot represents the base quality of the corresponding position along reads, color intensity reflects the number of nucleotides, a more intense color along a quality value indicates a higher proportion of this quality in the sequencing data.

4 References

[1] SOAPnuke: a MapReduce acceleration-supported software for integrated quality control and preprocessing of high-throughput sequencing data.

Chen Y, Chen Y, Shi C, et al.

PMID: 29220494 PMCID: PMC5788068 DOI: 10.1093/gigascience/gix120

Methods

1 Experimental procedure

The library construction method and sequencing process are carried out according to the following steps:

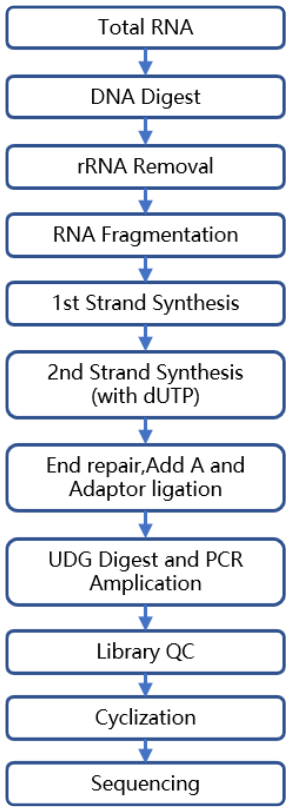


Figure 1 Workflow of experiment

1. DNA digest: total RNA was treated with DNase I which degraded double-stranded and single-stranded DNA presenting in RNA samples.

2. rRNA removal: using the Ribo-off rRNA Depletion Kit to remove ribosomal RNA.

3. RNA fragmentation.

4. cDNA Synthesis: First-strand cDNA was generated using random hexamer-primed reverse transcription, followed by a second-strand cDNA synthesis with dUTP instead of dTTP.

5. End repair, add A and adaptor ligation.

6. PCR: digest the U-labeled second-strand template with Uracil-DNA-Glycosylase (UDG) and perform PCR amplification.

7. Library quality control.

8. Cyclization: Denature the PCR product by heat and the DNA is cyclized by splint oligo and DNA ligase.

9. DNA nanoball (DNB) synthesis.

10. Sequencing on DNBSEQ platform.

2 Bioinformatic analysis workflow

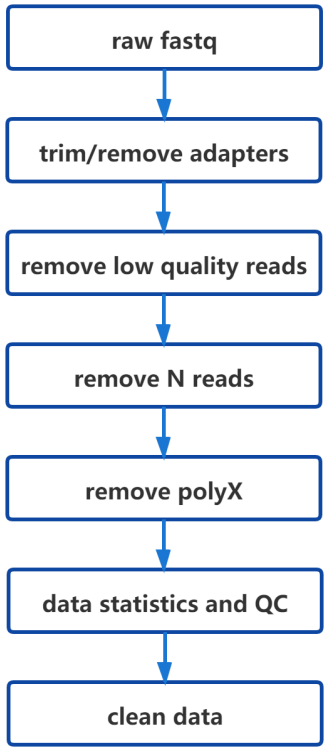


Figure 2 Bioinformatic analysis workflow

3 Parameters for data filtering

Raw data with adapter sequences or low-quality sequences was filtered. We first went through a series of data processing to remove contamination and obtain valid data. This step was completed by SOAPnuke software developed by BGI.

SOAPnuke software filter parameters: " -n 0.001 -l 20 -q 0.4 --adaMR 0.25 --polyX 50 --minReadLen 150", steps of filtering:

1. Filter adapter: if the sequencing read matches 25.0% or more of the adapter sequence (maximum 2 base mismatches are allowed), remove the entire read;
2. Filter read length: if the length of the sequencing read is less than 150 bp, discard the entire read;
3. Remove N: if the N content in the sequencing read accounts for 0.1% or more of the entire read, discard the entire read;
4. Remove polyX: if the length of polyX (X can be A, T, G or C) in the sequencing read exceeds 50bp, discard the entire read;
5. Filter low-quality data: if the bases with a quality value of less than 20 in the sequencing read account for 40.0% or more of the entire read, discard the entire read;

6. Obtain Clean reads: the output read quality value system is set to Phred+33.

4 References

[1] SOAPnuke: a MapReduce acceleration-supported software for integrated quality control and preprocessing of high-throughput sequencing data.

Chen Y, Chen Y, Shi C, et al.

PMID: 29220494 PMCID: PMC5788068 DOI: 10.1093/gigascience/gix120

Help

1 FASTQ format description

Images generated by sequencers are converted by base calling into nucleotide sequences, which are called raw data or raw reads and are stored in FASTQ format. FASTQ files are text files that store both reads sequences and their corresponding quality scores. Each read is described in four lines as follows:

```
@V350016857L4C001R0010000078/1
TTTTTCTGCTCCTTTTGATGCTATTAACAATTGCTTCAAGTTCAAGGGCACCTGCCTCAAAGTCCCTTTCTTCCAGACAA
+
=,DDE@EFFF=DFDEFCCFDEFEGFEEAFDFFE=FFCFFEEEDFDEEEFDF8FFEFFEFF:FFEDF=EFDGE<1FDCEFFFFFDFE
```

The first line is the sequence identifier and related description information, starting with'@'; the second line is the base sequence information; the third line starts with'+', followed by the sequence identifier, description information, or nothing; The four lines are quality information, which corresponds to the sequence in the second line. Each base has a quality score. Depending on the scoring system, each character represents a different quality value.

The figure below shows the concise correspondence between the sequencing error rate and the sequencing quality value. Specifically, if the sequencing error rate is denoted by E and the base quality value is denoted by SQ, there are the following relationships:

$$SQ = -10 \times (\log \frac{E}{1-E}) / (\log 10)$$

$$E = \frac{Y}{1+Y}$$

$$Y = \frac{SQ}{e^{-10 \times log 10}}$$

1. For the quality system data with a sequencing quality value of 33: the sequencing quality value of the base = the ASCII value corresponding to the quality information character -33, for example, the ASCII value corresponding to A is 65, then the corresponding base quality value is 65-33 = 32. The base quality value of the DNBSEQ sequencing platform ranges from 2 to 42.

2. For quality system data with a sequencing quality value of 64: the sequencing quality value of the base = the ASCII value corresponding to the quality information character -64, for example, the corresponding ASCII value of c is 99, then the corresponding base quality value is 99-64 =35. The base quality value of the DNBSEQ sequencing platform ranges from 2 to 43.

Table 1 A summary table of the relationship between sequencing error rate and sequencing quality

Sequencing error rate	Sequencing quality value	Character(Phred64)	Character(Phred33)
5%	13	M	.
1%	20	T	5
0.1%	30	^	?